

Module 5 — Read 1

Correlation, least-squares regression, residuals, and model performance

Module 5, Read 1: Correlation and the least-squares line

Project: [Project 5 — Looking Ahead](#) · **Preview:** [Read 0](#) · **Prerequisite:** [Module 3 scatterplots](#) · **Next:** [Quiz A \(Def & Notation\)](#), then [Read 2](#).

This reading is your **self-paced textbook** for Module 5, Part 1. You will learn to **quantify** a linear relationship with Pearson r and to **model** it with a least-squares regression line — the foundation of Project 5.

Learning outcomes

Work through the section below; use **Check yourself** boxes before [Quiz A](#).

After this reading you should be able to:

Section 1: Correlation and the least-squares line

Scatterplots, Pearson r , least-squares regression, residuals, and model performance

- Read a scatterplot using **form** (linear or curved), **direction** (positive or negative), and **strength** (strong, moderate, or weak).
- Compute and interpret **Pearson correlation** r as a measure of strength and direction of a linear association.
- Fit a **simple linear regression** line $\hat{y} = b_0 + b_1x$ with `lm()` and identify slope b_1 and intercept b_0 .
- Interpret the **slope** in context: predicted change in y for a one-unit increase in x .
- Use the fitted equation to make a **point prediction** for a new x value.
- Define a **residual** as observed minus predicted ($y - \hat{y}$) and explain what the least-squares criterion minimizes.
- Interpret **MSE** and R^2 on a test set as summaries of predictive performance on new data.

Section 1: Correlation and the least-squares line

Correlation and the least-squares line

i Bridge from Module 3

In **Module 3 Read 1** you learned to **describe** scatterplots by form, direction, and strength. Here we add a **numerical summary** (Pearson r) and a **predictive model** (the least-squares line $\hat{y} = b_0 + b_1x$).

From scatterplot to simple linear regression models

For two numerical variables, recall how we read a scatterplot:

- form (linear or curved or neither),
- direction (positive or negative or none),
- strength (tight or loose around a trend).

When the form looks roughly linear, we can summarize linear relationship with:

- Pearson correlation r , and
- a least-squares regression line.

In this section

Quantify and model a linear relationship between two numerical variables.

Pearson correlation and least-squares line

For a response y and explanatory variable x :

- Pearson Correlation (`cor`) is unitless and ranges from -1 to 1 and is a one number summary of the direction and strength of a linear relationship between two numerical variables (assumes that the form is linear)
- The fitted line is:

$$\hat{y} = b_0 + b_1x$$

where b_1 is slope and b_0 is intercept.

In R, fit with `lm(y ~ x, data = ...)`.

Optional activity: [visualize best-fit line choices](#) — try different lines and see MSE/MAE change.

Sample correlation formula and interpretation

For sample data (x_i, y_i) , $i = 1, \dots, n$, the sample correlation is

$$r = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i=1}^n (x_i - \bar{x})^2} \sqrt{\sum_{i=1}^n (y_i - \bar{y})^2}}$$

How to read this:

- **Numerator:** adds products $(x_i - \bar{x})(y_i - \bar{y})$.

- If x and y are often both above their means (or both below), products tend to be positive, so r tends to be positive (positive direction).
- If one is often above mean while the other is below mean, products tend to be negative, so r tends to be negative (negative direction).
- **Denominator:** rescales by overall spread in x and y , which keeps r between -1 and 1 .
- r near 0 means no clear **linear** direction.

Quick interpretation reminder:

- If $|r|$ is closer to 1 (for example -0.9 or 0.9), the linear relationship is stronger.
- If r is close to 0 , the linear relationship is weak to none.
- The sign gives direction:
 - $r > 0$: positive direction (as x increases, y tends to increase).
 - $r < 0$: negative direction (as x increases, y tends to decrease).

Practice activity: Rossman Chance “Guess the Correlation”

<https://www.rossmanchance.com/applets/guesscorrelation/GuessCorrelation.html>

Condition reminder for using correlation

Pearson correlation is a linear-association summary. Use it when the scatterplot form is roughly linear (not strongly curved and not dominated by outliers).

Example 1 - Study hours and exam score

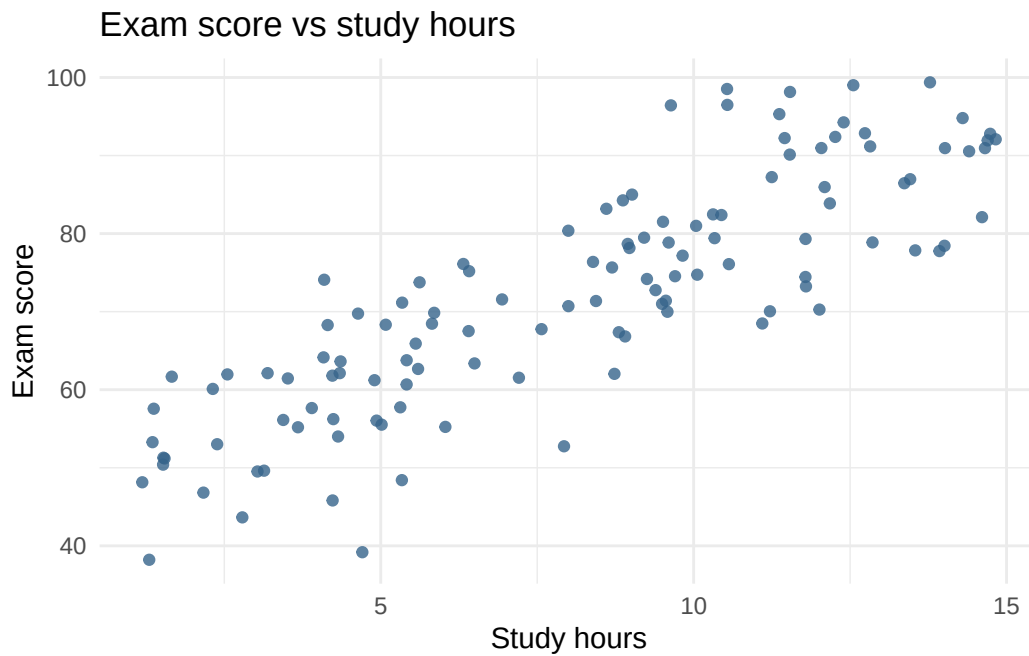
```
if (!requireNamespace("tidyverse", quietly = TRUE)) {
  install.packages("tidyverse", repos = "https://cloud.r-project.org")
}

library(tidyverse)

set.seed(3401)
n <- 120
exam_dat <- tibble(
  study_hours = runif(n, 1, 15),
  exam_score = 45 + 3.4 * study_hours + rnorm(n, 0, 8)
)
```

1) Scatterplot: form, direction, strength

```
ggplot(exam_dat, aes(x = study_hours, y = exam_score)) +
  geom_point(alpha = 0.8, color = "steelblue4") +
  labs(
    title = "Exam score vs study hours",
    x = "Study hours",
    y = "Exam score"
  ) +
  theme_minimal()
```



Interpretation: the relationship is roughly linear, positive, and moderate-to-strong.

2) Pearson correlation

```
cor(exam_dat$study_hours, exam_dat$exam_score)
```

```
[1] 0.8441325
```

3) Training and test split (80/20)

```
set.seed(3402)
idx_train <- sample(seq_len(nrow(exam_dat)), size = 0.8 * nrow(exam_dat))
exam_train <- exam_dat[idx_train, ]
exam_test  <- exam_dat[-idx_train, ]

nrow(exam_train)
```

```
[1] 96
```

```
nrow(exam_test)
```

```
[1] 24
```

4) Fit least-squares regression on training data

```
fit_exam <- lm(exam_score ~ study_hours, data = exam_train)
summary(fit_exam)
```

Call:

```
lm(formula = exam_score ~ study_hours, data = exam_train)
```

Residuals:

Min	1Q	Median	3Q	Max
-----	----	--------	----	-----

```
-19.1928 -4.7092 -0.2337 5.4140 16.5633
```

Coefficients:

```
      Estimate Std. Error t value Pr(>|t|)
(Intercept)  47.6895     1.7424   27.37  <2e-16 ***
study_hours   3.0594     0.1906   16.05  <2e-16 ***
---

```

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 7.519 on 94 degrees of freedom
Multiple R-squared: 0.7326, Adjusted R-squared: 0.7298
F-statistic: 257.5 on 1 and 94 DF, p-value: < 2.2e-16

```
coef(fit_exam)
```

```
(Intercept) study_hours
  47.689527    3.059362
```

Interpretation:

- slope b1: predicted increase in exam score is 3.06 points for each additional hour of study.
- intercept b0: predicted score is 47.7 when study hours = 0

```
new_hours <- tibble(study_hours = 10)
predict(fit_exam, newdata = new_hours)
```

```
1
78.28315
```

If a student studies 10 hours then our linear model predicts they will get an exam score of 78.3. By hand, $\hat{score} = 47.7 + 3.06(10)$.

Residuals and model performance: training vs test

Residual is observed y - predicted y .

```
exam_train <- exam_train |>
  mutate(
    pred = predict(fit_exam, newdata = exam_train),
    resid = exam_score - pred
  )

exam_test <- exam_test |>
  mutate(
    pred = predict(fit_exam, newdata = exam_test),
    resid = exam_score - pred
  )
```

Residual descriptives for both sets

```
resid_desc <- bind_rows(
  exam_train |>
    summarise(
```

```

    set = "train",
    n = n(),
    mean_resid = mean(resid),
    sd_resid = sd(resid),
    min_resid = min(resid),
    q1_resid = quantile(resid, 0.25),
    median_resid = median(resid),
    q3_resid = quantile(resid, 0.75),
    max_resid = max(resid)
  ),
exam_test |>
  summarise(
    set = "test",
    n = n(),
    mean_resid = mean(resid),
    sd_resid = sd(resid),
    min_resid = min(resid),
    q1_resid = quantile(resid, 0.25),
    median_resid = median(resid),
    q3_resid = quantile(resid, 0.75),
    max_resid = max(resid)
  )
)
resid_desc

```

```

# A tibble: 2 x 9
  set      n mean_resid sd_resid min_resid q1_resid median_resid q3_resid
<chr> <int>    <dbl>    <dbl>    <dbl>    <dbl>        <dbl>    <dbl>
1 train   96  1.78e-15    7.48   -19.2    -4.71     -0.234    5.41
2 test    24 -1.82e+ 0    9.61   -22.9    -6.87     -1.57     2.80
# i 1 more variable: max_resid <dbl>

```

MSE and R-squared on training and test

```

mse <- function(y, yhat) mean((y - yhat)^2)

mse_train <- mse(exam_train$exam_score, exam_train$pred)
mse_test  <- mse(exam_test$exam_score, exam_test$pred)

ss_tot_train <- sum((exam_train$exam_score - mean(exam_train$exam_score))^2)
ss_tot_test  <- sum((exam_test$exam_score - mean(exam_test$exam_score))^2)
ss_res_train <- sum((exam_train$exam_score - exam_train$pred)^2)
ss_res_test  <- sum((exam_test$exam_score - exam_test$pred)^2)

r2_train <- 1 - ss_res_train / ss_tot_train
r2_test  <- 1 - ss_res_test  / ss_tot_test

tibble(
  set = c("train", "test"),
  MSE = c(mse_train, mse_test),
  R2 = c(r2_train, r2_test)
)

```

```
)
```

```
# A tibble: 2 x 3
  set      MSE    R2
  <chr> <dbl> <dbl>
1 train  55.4 0.733
2 test   91.8 0.623
```

Interpretation notes:

- Training MSE is optimistically biased (it is measured on the same data used to fit the model).
- Test MSE is a less biased estimate of prediction error on new data.
- R^2 is a rescaled error summary: roughly, the proportion of variability in y explained by linear prediction from x .
- Test R^2 should be reported for expected out-of-sample performance.

Example 2 - Fertilizer dose and plant height

```
set.seed(3403)
plant_dat <- tibble(
  dose = runif(90, 0, 12),
  height_cm = 18 + 2.1 * dose + rnorm(90, 0, 5)
)

cor(plant_dat$dose, plant_dat$height_cm)
```

```
[1] 0.8321912
```

```
idx <- sample(seq_len(nrow(plant_dat)), size = 0.8 * nrow(plant_dat))
plant_train <- plant_dat[idx, ]
plant_test <- plant_dat[-idx, ]

fit_plant <- lm(height_cm ~ dose, data = plant_train)
coef(fit_plant)
```

```
(Intercept)      dose
  17.737388    2.359837
```

```
plant_test <- plant_test |>
  mutate(pred = predict(fit_plant, newdata = plant_test))

tibble(
  test_MSE = mse(plant_test$height_cm, plant_test$pred),
  test_R2 = 1 - sum((plant_test$height_cm - plant_test$pred)^2) /
    sum((plant_test$height_cm - mean(plant_test$height_cm))^2)
)
```

```
# A tibble: 1 x 2
  test_MSE test_R2
  <dbl>    <dbl>
1    25.7    0.651
```


Example 3 - Exercise minutes and resting heart rate

```
set.seed(3404)
heart_dat <- tibble(
  exercise_min = runif(100, 0, 60),
  resting_hr = 78 - 0.28 * exercise_min + rnorm(100, 0, 4.2)
)

cor(heart_dat$exercise_min, heart_dat$resting_hr)
```

```
[1] -0.780774
```

```
idx <- sample(seq_len(nrow(heart_dat)), size = 0.8 * nrow(heart_dat))
heart_train <- heart_dat[idx, ]
heart_test <- heart_dat[-idx, ]

fit_heart <- lm(resting_hr ~ exercise_min, data = heart_train)
coef(fit_heart)
```

```
(Intercept) exercise_min
  76.526712    -0.247786
```

```
heart_test <- heart_test |>
  mutate(pred = predict(fit_heart, newdata = heart_test))

tibble(
  test_MSE = mse(heart_test$resting_hr, heart_test$pred),
  test_R2 = 1 - sum((heart_test$resting_hr - heart_test$pred)^2) /
    sum((heart_test$resting_hr - mean(heart_test$resting_hr))^2)
)
```

```
# A tibble: 1 x 2
  test_MSE test_R2
    <dbl>   <dbl>
1    18.0    0.672
```

Check yourself — Correlation and the least-squares line

1. A scatterplot shows study hours on the x-axis and exam score on the y-axis. Points slope upward and cluster near a line. Describe form, direction, and strength.
2. Pearson $r = -0.82$ between exercise minutes and resting heart rate. Interpret this value.
3. R output: (Intercept) 47.7, study_hours 3.06. Write the prediction equation and interpret the slope.
4. Using the equation above, predict exam score for a student who studies 10 hours. Show the arithmetic.
5. Define **residual**. For a student with exam score 85 who studied 10 hours, compute the residual.
6. Why should you prefer **test** MSE and R^2 over training metrics when reporting model performance?

Answers — Correlation and the least-squares line

1. **Form:** roughly linear. **Direction:** positive (more hours $\rightarrow$ higher score). **Strength:** moderate to strong.
2. Strong **negative** linear association: as daily exercise minutes increase, resting heart rate tends to decrease.
3. $\hat{y} = 47.7 + 3.06x$. **Slope:** each additional hour of study is associated with a predicted 3.06-point increase in exam score.
4. $\hat{y} = 47.7 + 3.06(10) = 47.7 + 30.6 = 78.3$ points.
5. Residual $= y - \hat{y} = 85 - 78.3 = 6.7$ — the student scored 6.7 points **above** the model prediction.
6. Training metrics are optimistically biased — the model was fit on the same data. Test metrics estimate performance on **new data** the model has not seen.

i Optional enrichment (not required for Project 5)

- **Logistic regression** — modeling a binary response with `glm()` and odds ratios